χ-Theory: Control-Dependent Spectral Non-Equivalence

Abstract

We show that identical Hamiltonian dynamics do not imply equivalence of physical observables when control-dependent measurement statistics are taken into account. We introduce an operational distinguishability structure W defined by control-conditioned outcome distributions $P(x|u)$, and construct a reconstructed operator $\hat{\Delta}_\chi$ whose spectrum defines physical observables. We prove that different control implementations of the same Hamiltonian generically lead to unitarily inequivalent operators and therefore different spectra.

1 Operational Setting

Let $\mathcal{H}$ be a separable Hilbert space and $H \in \mathcal{B}(\mathcal{H})$ a bounded Hamiltonian generating unitary evolution:

$$U(t) = e^{-iHt}.$$

Let $\mathcal{U}$ denote the set of admissible control protocols $u(t)$.

For each $u \in \mathcal{U}$, measurement outcomes are described by POVMs $\{E_x^{(u)}\}$, giving probabilities:

$$P(x|u) = \mathrm{Tr}(\rho_u E_x^{(u)}).$$

Define the operational distinguishability structure:

$$W := \{P(x|u)\}_{u \in \mathcal{U}}.$$

2 Tomographic Reconstruction

Assume the existence of a reconstruction map:

$$\mathcal{T} : W \to \mathcal{B}(\mathcal{H}), \qquad \hat{\Delta}_\chi := \mathcal{T}(W),$$

well-defined up to unitary equivalence under informational completeness of the control family.

3 Central Postulate

$$\boxed{\mathcal{O}_{phys} = \mathrm{Spec}(\hat{\Delta}_\chi)}$$

Physical observables correspond to the spectrum of the reconstructed χ-operator.

4 Main Theorem

Control-Dependent Spectral Non-Equivalence

Let two implementations satisfy:

$$H^{(1)} = H^{(2)}.$$

Let their operational structures be:

$$W^{(1)} = \{P_1(x|u)\}, \quad W^{(2)} = \{P_2(x|u)\}.$$

If:

$$W^{(1)} \neq W^{(2)},$$

then generically:

$$\boxed{\mathrm{Spec}(\hat{\Delta}_\chi^{(1)}) \neq \mathrm{Spec}(\hat{\Delta}_\chi^{(2)})}$$

where:

$$\hat{\Delta}_\chi^{(k)} = \mathcal{T}(W^{(k)}).$$

5 Falsification Criterion

The theory is falsified if there exist distinct operational structures:

$$W^{(1)} \neq W^{(2)}$$

such that:

$$\mathrm{Spec}(\hat{\Delta}_\chi^{(1)}) = \mathrm{Spec}(\hat{\Delta}_\chi^{(2)})$$

within experimental precision for informationally complete control sets.

6 Experimental Relevance

The framework is testable in systems where identical Hamiltonians can be implemented via different control protocols, including:

- trapped-ion quantum systems
- superconducting qubits
- programmable quantum simulators

7 Main Result

$$\boxed{H^{(1)} = H^{(2)} \;\not\Rightarrow\; \text{physical equivalence}}$$

Physical equivalence additionally requires equality of control-induced operational structures:

$$W^{(1)} = W^{(2)}.$$